

Aly Gaber

aly.gaber@mail.mcgill.ca | [linkedin.com/in/aly-gaber](https://www.linkedin.com/in/aly-gaber) | github.com/AlyGaber0 | alygaber-dev.vercel.app | +1 (263) 380-8533

EDUCATION

McGill University, Montreal, QC

September 2024 – May 2028 (Expected)

Bachelor of Engineering in Software Engineering, Co-op, Minor in Mathematics

- *Systems Coursework:* Computer Organization, Computer Architecture (W27), Compiler Design (W27), Digital Logic, Intro to Software Systems (C, Unix), Signals and Networks
- *Other Coursework:* Data Structures and Algorithms, Software Engineering (Java, OOP), Probability and Statistics, Linear Algebra I & II, Real Analysis

PROJECTS

Gabl | *Interpreted Language in C* | alygaber0.github.io/Gabl

May 2026 – August 2026

- Built an interpreted language in C with no parsing libraries: hand-written lexer, recursive descent parser encoding operator precedence as tree depth, and a tree-walking AST evaluator
- Represented runtime values as a tagged union, redesigning its layout when strings and arrays were added as composite types
- Implemented lexically scoped environments with parent-chain lookup, isolating state across recursive and nested calls; desugared for loops in the parser
- Traced a pointer-truncation segfault and a scope leak between nested calls; added a test suite, Makefile build, and documentation site

Coursework Projects | *McGill University*

2025 – Present

- *Network protocols:* Implemented a record protocol over TCP sockets in Python: length-prefixed frames with CRC-32 validation, loops handling partial writes and short reads, and a DATA/ACK/CLOSE state machine
- *Intro to software engineering:* Led a seven-person team building a retail platform in Java and Spring Boot on a layered architecture (model, repository, service, controller, DTO), with a Vue.js frontend and separate access for customer, employee, and manager roles
- *Computer organization:* Built a working CPU in Logisim-Evolution: datapath and control for a small instruction set
- *Digital logic:* Designed combinational and sequential VHDL circuits (adders, clock divider, Moore FSM sequence detector), verified in ModelSim and synthesized to a DE1-SoC FPGA in Quartus, with timing analysis for critical-path delay

EXPERIENCE

Vexo Labs Inc. (fintech), Remote

January 2026 – April 2026

Software Engineering Intern

- Built four customer-facing pages end to end at an early-stage fintech, including a live market dashboard rendering real-time prices across a broad ticker list alongside the day's top three gainers and losers (Next.js, TypeScript)
- Integrated those pages with a Supabase (PostgreSQL) backend, writing the queries behind each view against the team's schema

Darwinz AI (TheDar.AI), Cairo, EG

May 2025 – July 2025

Software Engineering Intern

- Built a document pipeline on GCP (Cloud Functions, Vertex AI) that ran PDFs through OCR, translation, and summarization and loaded structured metadata into BigQuery for SQL analysis
- Wrote and deployed the project's Flask REST services on Google Cloud, containerized with Docker

TECHNICAL SKILLS

Languages: C, C++, Python, Java, SQL, TypeScript, JavaScript, ARM Assembly, VHDL

Tools and Infrastructure: Git, Make, Docker, GCP (Cloud Functions, Vertex AI), PostgreSQL, BigQuery

Web and Frameworks: React, Next.js, Node.js, Spring Boot, Flask, Supabase, REST APIs

EXTRACURRICULARS

McGill Squash Team | Men's B National Champion, undefeated record

September 2024 – Present